GG-M-100a

MARCH 24, 1958

SUPERSEDING
Int. Fed. Spec. GG–M–00100 (GSA–FSS)
August 15, 1956, and
Fed. Spec. GG–L–661
December 6, 1946

FEDERAL SPECIFICATION

MAGNIFIER (BINOCULAR)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers binocular magnifiers for magnification of work leaving both hands free.

1.2 Classification.

1.2.1 *Types.*—Binocular magnifiers, in accordance with this specification, shall be of the following types, as specified (see 6.1).

Type I. —Beebe.

Type II.—Berger.

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specifications and standards of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

GG–I–526 — Instruments, Dental and Surgical; General Specifications.

PPP–B–621—Boxes, Wood, Nailed and Lock-Corner.

PPP–B–636—Boxes, Fiber.

PPP–B–676 — Boxes, Set-Up, Paperboard.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL–L–10547 — Liners, Case, Waterproof.

Military Standards:

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129 — Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 **Material.** — Except where otherwise specified, metal parts for type I magnifiers shall be fabricated of nickel silver, chromium over nickel plated. Plastic for type II magnifier shall be a molded black plastic.

3.2 **Marking.** — Each binocular magnifier shall be permanently and legibly marked with the name or registered trade-mark of the manufacturer.

3.3 **Type I.**—Type I magnifier shall be of the Beebe type. The general design shall be as illustrated in figure 1.

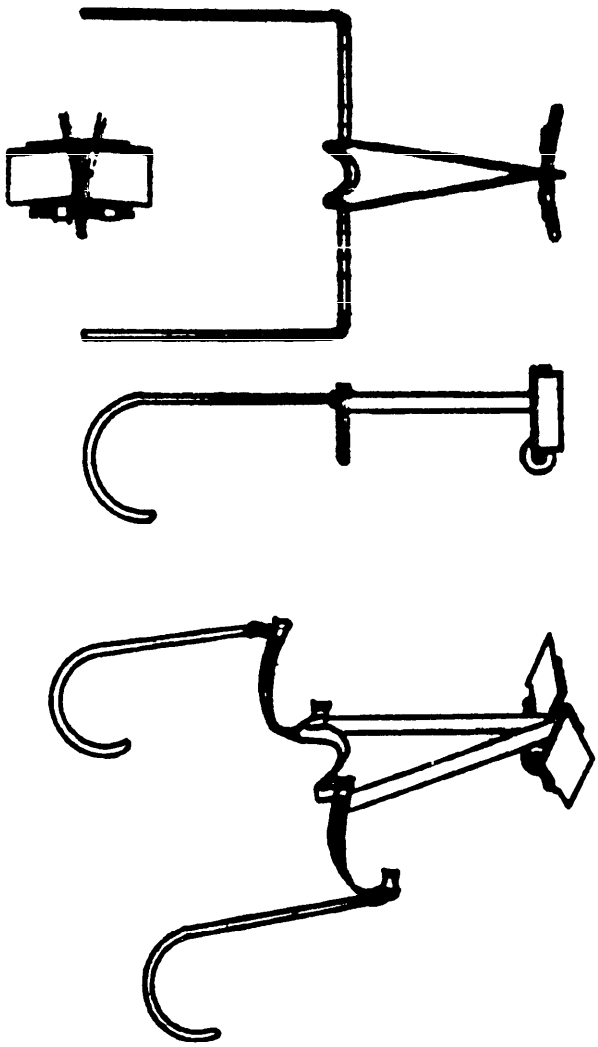


FIGURE 1.—Type I, Beebe.

3.3.1 *Frame.*—The frame shall consist of two trial lens cells connected by a nasal bridge, each cell capable of receiving a 1.50-inch trial lens, and two adjustable flexible wire temple pieces approximately 6.5 plus or minus 0.5 inches long, firmly hinged to the frame by metal screws.

3.3.1.1 *Temples.* — The temples shall be constructed of a single flexible wire, with or without a flexible cable connected at the bowed (ear) end, and bowed at one end to fit comfortably behind the ear. A vinyl tubing may be used to cover the entire length of the temple, or the bowed portion that contacts the ear, in which case the nickel and chromium plating of the wire may be omitted.

3.3.2 *Magnifier attachment.*—The magnifier attachment shall be integral with the lens frame and shall consist of two pieces of metal 0.250 inch in width, 0.031 inch in thickness, and of sufficient length to mount two lens prisms 76 millimeters in front of the lens cells. Each lens prism shall be an 8.00-diopter sphere with a 5-degree prism. The lens shall measure a minimum of 21 millimeters (mm.) by 25 millimeters. They shall be mounted on a leaded brass prism tube, base in, and shall be so arranged that the interpupillary space may be adjusted by means of a knurled thumbscrew. Manipulation of this thumbscrew shall move lenses for both eyes simultaneously and synchronously, but in opposite directions, in the horizontal plane. The thumbscrew shall be capable of adjusting the interpupillary distance with a minimum range of from 30 millimeters to 40 millimeters.

3.3.3 *Plating.*—All metal parts shall have a composite plating of chromium over nickel; minimum thicknesses of these layers at any point on significant surfaces shall be respectively as follows:

Chromium	0.00001 inch.
Nickel	0.0001 inch.

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3.4 **Type II.**—Type II magnifiers shall be of the Berger type. The general design shall be as illustrated in figure 2. The magnifiers shall be made from all-black, lightweight plastic. The magnifiers shall be adjustable to interpupillary distance by means of a connecting hinge at the distal end of the barrels, and to facial contour by means of a swivel joint connecting the barrels and facial cups.

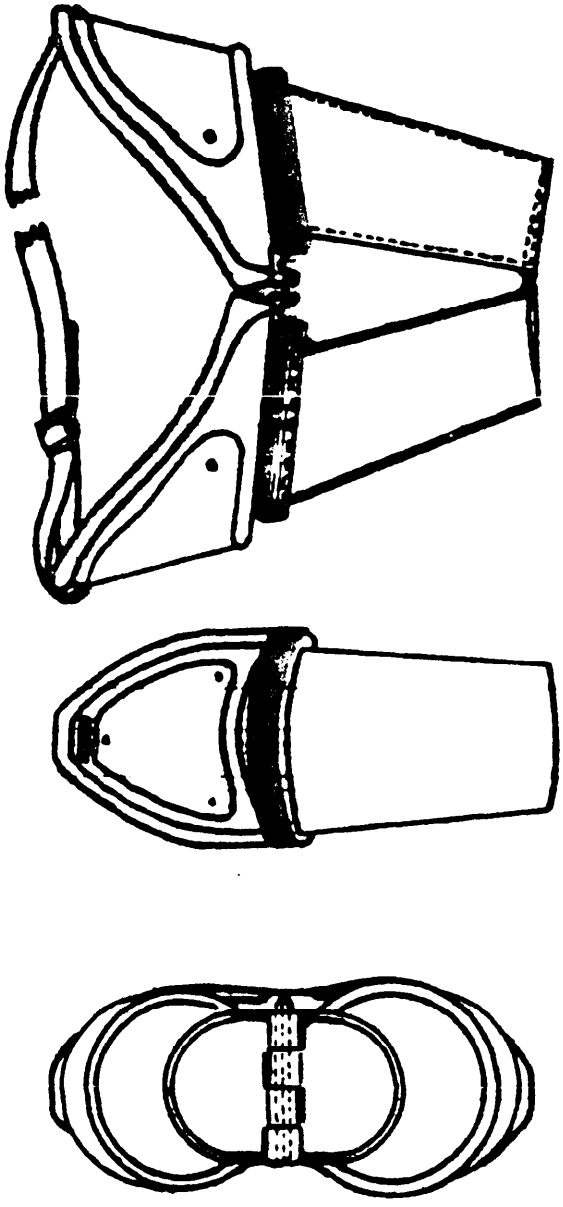


FIGURE 2.—Type II, Berger.

3.4.1 *Over-all dimensions.*—The over-all dimensions shall be not greater than the following:

Length	5	inches.
Width	5.25	inches.
Depth	2.25	inches.

3.4.2 *Lens.*—The lens prism shall be a 10.00-diopter biconvex sphere with a 5.5-degree prism. The lens shall be 38 millimeters in diameter, cut down to 27 millimeters on one side to form the prism base, and shall be mounted in a base-in position.

3.4.3 *Head band.*—The magnifier shall be held in place by a black elastic band not less than 0.50 inch in width and approximately 21 inches in length. The band shall be provided with an adjustable metal side fastener.

3.4.4 *Nasal-bridge pad.*—The nasal-bridge pad shall be made of soft leather cemented to a felt lining. The pad shall be approximately 1 inch in depth by 1.125 inch at its widest point, and shaped to fit the bridge of the magnifier without interference with vision. The pad shall have suitable straps of the same material for attachment and an adjustable metal fastener. The pad shall be attached so that the felt side will come in contact with the wearer.

3.4.5 *Finish.*—All parts shall be smoothly and properly polished.

3.5 **Workmanship.**—The workmanship shall be first class throughout. The magnifiers shall be smoothly and properly finished and free from all defects which affect their appearance or may affect their serviceability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 **Sampling.**—Sampling procedures shall be conducted in accordance with procedures set forth in Military Standard MIL–STD–105. In addition, samples may be requested by the procuring activity for testing at Government laboratory.

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4.2 **Inspection.**—Inspection may be made throughout the entire process of manufacture. The passing as satisfactory of any detail construction or materials shall not relieve the contractor of responsibility for faulty workmanship or materials which may be discovered at any time prior to final acceptance. Final inspection of the finished article shall be made either at point of production or at point of delivery as specified (see 6.1). In case of factory inspection, every facility shall be afforded inspectors, by the manufacturer, for the prosecution of their work.

4.3 Tests.—Plating tests shall be conducted in accordance with the applicable requirements of Federal Specification GG–I–526.

5. PREPARATION FOR DELIVERY

(The levels of Preservation, Packaging and Packing protection used are defined in Fed. Std. No. 102.)

5.1 **Packaging.**

5.1.1 *Unit of issue.*—One (1) magnifier, as specified, constitutes one unit.

5.1.2 *Packaging quantities.*—The number of units of issue indicated in table I shall be packaged in each unit and exterior container, as applicable, for the required level of protection specified in procurement document.

TABLE I.—*Packaging quantities*

Federal Stock Number	Unit package	Intermediate package	Shipping container
6515–346–5900	1 unit	Not required	24 units

5.1.2.1 *Packing variation permitted.*—If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size, and design, acceptable to a common carrier, which will insure safe delivery to destination.

5.1.3 *Level A.*

5.1.3.1 *Unit package.*—Each unit shall be packaged in a carton of appropriate size constructed in accordance with Federal Specification PPP–B–636, type I, class 1, or PPP–B–676. Outside dimensions of carton shall not exceed 5½ inches by 3 13/16 inches by 3⅛ inches. Carton shall be designed with suitable interior fittings or shall contain sufficient cushioning material to insure against shifting of, or damage to, the contents. Each carton shall be adequately secured to prevent accidental opening.

5.1.4 *Level C.*—Units shall be packaged in standard commercial containers of the size, and kind, commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose within the continental limits of the United States of America.

5.2 Packing.

5.2.1 *Level A.*

5.2.1.1 *Exterior container.*—Exterior container shall be constructed in accordance with Federal Specification PPP–B–621, table IV.

5.2.1.2 *Case liner.*—Each level A box shall be lined with a waterproof case liner conforming to Military Specification MIL–L–10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto.

5.2.1.3 *Strapping.*—Strapping, when required, shall be in accordance with applicable box specification.

Note.—Strapping will not be required for shipment forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 *Level B.*

5.2.2.1 *Exterior container.*—Exterior container shall be constructed in accordance with Federal Specification PPP–B–621, table II.

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5.2.3 *Level C.* — The subject commodity shall be packed in substantial commercial containers of the type, size, and kind, commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 **Marking.**

5.3.1 *Unit package.*—Each unit package shall be marked in accordance with section 5 of Military Standard MIL–STD–129.

5.3.2 *Shipping container.*—Shipping container shall be marked in accordance with Military Standard MIL–STD–129. Service color and service symbol markings are required only on direct shipments from the contractor's plant to overseas activities.

6. NOTES

6.1 **Ordering data.** — Purchasers should specify type required and should exercise any desired options offered herein. (See 1.2.1, 4.2, 5.1 and 5.2.)

6.2 It is believed that this specification adequately describes the characteristics necessary to secure the desired material, and that normally no samples will be necessary prior to award to determine compliance with this specification. If, for any particular purpose, samples with bids are necessary, they should be specifically asked for in the invitation for bids, and the particular purpose to be served by the bid sample should be definitely stated, the specification to apply in all other respects.

6.3 This specification does not include all types, classes, grades, etc., of the commodity indicated by the title of the specification, or which are commercially available, but is intended to cover the type, etc., normally procured to meet Federal Government requirements.

6.4 **Transportation description.**—Transportation description applicable to this item is:

Optical goods, not otherwise indexed by name.

Carload minimum weight 10,000 pounds, subject to Rule 34, Uniform Freight Classification.

Motor volume minimum weight 10,000 pounds.

6.5 This specification covers the following item appearing in the Armed Services Medical Stock List:

6515–346–5900 MAGNIFIER, Binocular, Spectacle Design.

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTERESTS:

Army—M
Navy—Md
Air Force.

Copies of this specification may be purchased for 5 cents each.

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